



Roots of *Cynodon dactylon* increase gas permeability and gas diffusion coefficient of highly compacted soils

S. Feng · S. F. Huang · C. W. W. Ng · F. Q. Chen ·
X. Qian · N. K. Zhao

Received: 29 December 2022 / Accepted: 13 July 2023 / Published online: 14 August 2023
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2023

Abstract

Aims This study experimentally investigated the short-term (about 6 months) effects of roots of *Cynodon dactylon* on the gas permeability and gas diffusion coefficient of soils with different degrees of compaction.

Methods Compacted soil planted with *Cynodon dactylon* were left outdoors for about 6 months for plant growth. The measurements of rooted and bare soils included gas permeability, gas diffusion coefficient, root characteristics and soil microstructure. The relative effects of different root characteristic

parameters on gas permeability and gas diffusion coefficient were compared through grey relational analysis.

Results The root volume ratio had a greater effect on gas permeability and gas diffusion coefficient, compared with root area index, root area ratio, root length density, and root biomass ratio. When the degree of compaction $\geq 85\%$ (porosity ≤ 0.41 , bulk density $\geq 1.56 \text{ g cm}^{-3}$), the macro-pores at the root–soil interface increased gas permeability and gas diffusion coefficient, while negligible effects of roots on gas movement existed under degree of compaction of 80%. The increase in gas permeability by roots was more significant than that in gas diffusion coefficient. However, roots' increase of gas movement generally decreased at higher root volume ratio due to roots-occupied soil pores. Finally, gas permeability and gas

Responsible Editor: Al Imran Malik.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11104-023-06173-6>.

S. Feng · S. F. Huang · F. Q. Chen (✉)
College of Civil Engineering, Fuzhou University,
Fuzhou City, Fujian Province, China
e-mail: phdchen@fzu.edu.cn

S. Feng
e-mail: sfengaa@connect.ust.hk

S. F. Huang
e-mail: 1061191618@qq.com

C. W. W. Ng
Department of Civil and Environmental Engineering,
Hong Kong University of Science and Technology, Clear
Water Bay, Kowloon, Hong Kong
e-mail: cecwwng@ust.hk

X. Qian (✉)
College of Life Sciences, Fujian Agriculture and Forestry
University, Fuzhou City, Fujian Province, China
e-mail: qxxb2006@163.com

N. K. Zhao
China Energy Engineering Group Anhui Electric Power
Design Institute Co., LTD., Hefei City, Anhui Province,
China
e-mail: 2498422748@qq.com

diffusion coefficient of rooted soil were well predicted by newly-developed empirical models considering the effect of root volume ratio.

Conclusions Macro-pores at the root–soil interface tended to increase gas permeability and gas diffusion coefficient of soil with a degree of compaction $\geq 85\%$, while it is the opposite for root volume ratio.

Keywords Root-soil interaction · Root characteristics · Soil microstructure · Gas movement · Unsaturated soil

Abbreviations

A	Cross-sectional area of a soil sample (0.005 m ²)
$A(\alpha_1)$	$A(\alpha_1) = -\frac{1}{\beta^2} - \alpha_1^2$
$B(\alpha_1)$	$B(\alpha_1) = \alpha_1^4 \beta + \alpha_1^2 \left(\frac{2}{\beta} + 1 \right) + \frac{1}{\beta^3} + \frac{1}{\beta^2}$
C_0	Initial volumetric concentration of O ₂ (m ³ /m ³) in the active chamber before the test
C_i	O ₂ concentration in the active chamber after time t (m ³ /m ³)
D_0	Gas diffusion coefficient in free air (2.0 × 10 ⁻⁵ m ² /s for O ₂ at 20 °C)
D_p	Gas diffusion coefficient (m ² /s)
d	Average particle diameter of the pore matrix ($d_{50} = 0.7$ mm)
$G(R_v)$	A function fitted according to the test data of rooted soil in Eq. (13)
$g(R_v)$	A function fitted according to the test data of rooted soil in Eq. (12)
H	Height of the active diffusion chamber (100 mm)
k_a	Gas permeability (m ²)
L	Half of the height of a soil sample (10 mm)
Q	Rate of air flow out of the active chamber (m ³ /s)
R_b	Root biomass ratio
R_e	Reynolds number of air flow
R_v	Root volume ratio
t	Measurement duration (s)
v	Fluid velocity (m/s)
X_a	A dimensionless constant representing the effects of water blockage on gas movement in Eq. (8)
X_r	A function of a root characteristic parameter, $X_r = F(R_v)$
X'	An empirical factor considering pore tortuosity in Eq. (11)

X'_r	A function of a root characteristic parameter, $X'_r = G(R_v)$, see Eq. (17).
x	Height of a soil sample (0.02 m)
α_a	A constant reflecting the connectivity and tortuosity of soil pores (m ² ; see Eq. 8)
α_r	A function of a root characteristic parameter, $\alpha_r = f(R_v)$ (see Eq. 9)
α_1^2	$\alpha_1^2 = \frac{1}{\beta} - \frac{1}{3\beta^2} + \frac{4}{45\beta^3} + \frac{16}{945\beta^4}$
β	$\beta = \frac{H}{L\epsilon}$
ϵ	Soil gas content (the volume of gas in a unit volume of soil; m ³ /m ³)
μ	Dynamic viscosity of gas (Pa · s)
γ	Kinematic viscosity of fluid (equal to 1.48 × 10 ⁻⁵ m ² /s for air at 20 °C and an atmospheric pressure of 101.3 kPa)
ΔP	Gas pressure measured in the passive chamber (Pa)
ϵ_{th}	Percolation threshold for bare soil, below which gas cannot diffuse via the gas phase in the soil
$\epsilon_{th,r}$	Percolation threshold for rooted soil

Introduction

Understanding gas movement in unsaturated soil is crucial for better understanding of plant-soil interactions, as it can: (i) improve soil aeration and hence increase plant growth and crop yields (Gilman et al. 1982; Zhang et al. 2022); (ii) provide guidance on remediation of soil–plant systems contaminated with volatile organic compounds and help design more effective remediation strategies (Mäki et al. 2019); (iii) facilitate quantification of carbon sequestration and greenhouse gas emissions in grassland (Townsend-Small and Czimczik 2010). Gas movement in soil mainly relies on advection and diffusion, which are quantified in terms of gas permeability (k_a) and a gas diffusion coefficient (D_p), respectively. The k_a and D_p of a soil directly affect gas movement in the soil, thereby influencing the activity of soil microorganisms and the aerobic respiration of roots (Bian et al. 2020; Lipiec et al. 2012).

Soil microstructure is affected by soil compaction, soil moisture content, and soil type (Berisso et al. 2013; Chen et al. 2021; Nguyen et al. 2021; Liu and Feng 2022) and also by roots (Hu et al. 2019; Rasse

and Smucker 1998; Uteau et al. 2013). Soil macropores could be generated by roots, due to (i) the limited lifespan of roots and related natural turnover, wherein a proportion of roots senesce, shrink, and break down (Kamchoom et al. 2023); (ii) root shrinkage–swelling during drying–wetting cycles (Carminati et al. 2013; Nobel and Cui 1992; Uteau et al. 2013; Zhu et al. 2022a, b), and (iii) root decay upon external factors, such as burning or usage of herbicide (Kamchoom et al. 2022a, b). These processes collectively influence the movement of both water and gases within the soil (Jotisankasa and Sirirattanachai 2017; Luo et al. 2019; Ni and Ng 2019; Zhan et al. 2016). Researchers have investigated the effects of roots on the hydraulic properties of soil, such as its water retention capacity (Boldrin et al. 2017; Leung et al. 2015; Ng et al. 2016, 2022a) and hydraulic conductivity (Liu et al. 2018; Ng et al. 2022b; Nobel and Cui 1992; Song et al. 2017, among many others). However, few studies have examined the effects of roots on soil k_a (Ni and Ng 2019; Zhan et al. 2016) and D_p (Uteau et al. 2013). Zhan et al. (2016) measured the k_a of loess horizons before and after plants were added to soil and observed that the filling of soil pores by plant roots reduced k_a by an order of magnitude. Ni and Ng (2019) found that the k_a of root-bearing soil was lower than that of bare soil after 6 months of vegetative growth, while this trend was reversed after 15 months of vegetative growth. It seems the impact of root on k_a mainly depends on root-induced change in soil microstructure. When root increases macropores in soil, k_a of soil would increase. However, Zhan et al. (2016) and Ni and Ng (2019) have focused on only k_a and ignored D_p . Uteau et al. (2013) investigated the effects of roots on k_a and D_p but only examined loosely compacted soil used for agriculture; thus, their results are not relevant to high-compaction soil, which could be caused by agriculture vehicles. Moreover, plants exhibit different root characteristics (Ng et al. 2015), which can be depicted by various indices, such as root volume ratio (R_v , the volume of root per unit soil volume), root length density (root length per unit soil volume), and root area ratio (the ratio of cross-sectional area of roots to that of soil at a given depth). However, the effects of root characteristics on soil k_a and D_p remain unknown.

Soil compaction leads to agricultural land degradation and hence a decrease in crop yields (Unger and Kaspar 1994). A higher soil compaction could be

caused by usage of heavy farm machinery and intensive cropping (Chen et al. 2014). A higher soil compaction would lead to a decrease in soil porosity and macropores (Schaffer et al. 2007), thus reducing soil aeration and limiting root growth (Unger and Kaspar 1994). Meanwhile, roots could also alleviate soil compaction as roots occupied soil pores and changed soil pore size distributions. It has been found that plants (e.g., soybean; Yue et al. 2021) alleviated soil compaction through generation of macropores. Moreover, the alleviation effects depended on plant species and root characteristics. The vertically growing root was more effective in alleviating soil compaction than the laterally growing one (Yue et al. 2021). Ren et al. (2019) reported that tap roots penetrated compacted soil more effectively than the fibrous roots. This root-induced change in soil microstructure and compaction subsequently affected k_a and D_p of soil, thereby affecting soil aeration. Chen et al. (2014) reported that roots increased gas permeability of compacted clayey soil under high degree of compaction, while effects of roots on gas permeability of soil became negligible under low degree of compaction. They further found that effects of roots on gas permeability of soil depended on soil texture, plant species and root characteristics. However, effects of root characteristics on D_p in soils with different degrees of compaction still remain unknown, even though it is well known that oxygen consumed by root respiration mainly enters the soil through diffusion from the atmosphere (Ikeda and Nakamura 1996).

The present study investigated the effects of the roots of Bermuda grass (*Cynodon dactylon*) on the k_a and D_p of soils with different degrees of compaction. We hypothesized that roots affect k_a and D_p of soil through two mechanisms: (i) macropores along the root–soil interface that provide preferential flow paths for gas, leading to an increase in k_a and D_p ; and (ii) root-filled soil pores that impede gas movement, resulting in a decrease in k_a and D_p . The net effect of these mechanisms on k_a and D_p depends on various factors, including different root characteristic parameters, soil structure, and soil moisture content. We further hypothesized that effects of roots on k_a and D_p also depended on soil degree of compaction, which affects the microstructure of soil matrix (e.g., pore size distributions, pore connectivity) and root growth. Accordingly, a series of tests were conducted to measure k_a and D_p of soils, and then root characteristics were determined. The characteristics of soil pores

and the root–soil interface structure were analysed via scanning electron microscopy, while pore size distributions were measured via mercury intrusion porosimetry. Finally, empirical prediction models of k_a and D_p were established that consider the effects of roots.

Materials and methods

Soil and grass characteristics

The soil was categorized as silty sand according to Unified Soil Classification System (USCS; ASTM D2487-17 (2017a)). Before testing, the soil was screened through a 2 mm sieve to remove roots and large particles. The basic properties of the soil are presented in Table 1 (Chen et al. 2020). Bermuda grass was chosen as the study plant as it is a common perennial herb species in the tropical and subtropical regions of Asia, including southern China, Malaysia, and Thailand, and is frequently planted in landfill covers and slopes for ecological restoration and aesthetic reasons (Kamchoom et al. 2022a).

Test procedure

The silty sand was sprayed with water mist to achieve an optimal water content of 17.1% that was

determined by the Proctor compaction test. This moist soil was stored in a sealed plastic bag for at least 48 h to ensure moisture equalisation was achieved. Subsequently, the moist soil was compacted in layers, each with a thickness of 30 mm, in steel drums with an internal diameter of 330 mm and a height of 220 mm. The target degrees of compaction were 80%, 85%, 90%, and 95%, corresponding to porosities of 0.45, 0.41, 0.38, and 0.34, respectively. Two steel drums were prepared for each target degree of compaction: one drum contained soil planted with *C. dactylon*, and the other drum contained uncultivated soil (bare soil) that served as a reference. Thus, overall, eight steel drums with compacted soil were prepared. After soil compaction, *C. dactylon* turfs with similar characteristics were transplanted into four drums containing soils with different degrees of compaction. All of the drums were left outdoors to allow *C. dactylon* to grow for approximately 6 months; in contrast, the uncultivated drums were covered with geonet with an aperture of ~1 mm to prevent *C. dactylon* growth.

After this time, the ring-knife was used to collect intact soil samples at different depths (40, 60, 80, and 100 mm) from the drums. Before soil sample collection, the inner wall of the ring-knife was pasted with a thin layer of Vaseline to mitigate preferential gas flow along the interface between the ring-knife and soil. At least three soil samples

Table 1 Basic properties of the test soil

Properties	Measured value	Reference
Soil type	Silty sand	ASTM D2487-17 (2017a)
Specific gravity	2.66	ASTM D854 (2014)
Particle size distribution (%)		ASTM D422 (2007)
Coarse sand (2 – 4.75 mm)	0	
Medium sand (0.425 – 2 mm)	53.6	
Fine sand (0.075 – 0.425 mm)	26.1	
Silt (0.002 mm– 0.075 mm)	15.9	
Clay (<0.002 mm)	4.4	
Average particle size (d_{50})	0.7	
Atterberg limits		ASTM D4318 (2017b)
Liquid limit (LL) (%)	37.8	
Plastic limit (PL) (%)	28.2	
Plastic Index (PI)	9.6	
Optimum water content (%) ^a	17.12	ASTM D698 (2010)
Maximum dry density (g cm^{-3})	1.84	ASTM D698 (2010)

a. Optimum water content refers to the gravimetric water content at which the maximum dry density of soil is achieved during the Proctor compaction test

were collected from the same depth in each compacted soil drum. Extra soil was collected at each sampling depth to measure the gravimetric water content, which ranged from 15 to 25% (the associated range of degree of saturation was 35% to 55%). Each intact collected soil sample was immediately installed in the apparatus (Fig. 1) for D_p and k_a measurements, which were performed as described below (see Section "A unified apparatus for measuring the k_a and D_p of unsaturated soil"). Thereafter, the soil sample was uninstalled, and the top and bottom surfaces of the sample were each sprayed with 2 to 5 g of water mist to increase the soil water content by 2% to 5%. Then, the soil sample was stored in a sealed plastic bag for at least 48 h for moisture equalisation (Jotisankasa and Sirirattanachai 2017; Tang et al. 2011). Afterward, the k_a and D_p of the wetted soil sample was measured. This above procedure was performed using soils with various water contents, which enabled investigation of the influence of soil gas content ε on the k_a and D_p of this soil. Soil gas content were calculated according to the measured soil water contents and specific gravity of soil particles. Then, soil specimens were collected to analyse soil microstructure and measure root characteristics.

A unified apparatus for measuring the k_a and D_p of unsaturated soil

Figure 1 shows a unified apparatus for measuring the k_a and D_p of unsaturated soil. The apparatus was made of polymethyl methacrylate and mainly consisted of three chambers: a passive chamber, an active chamber, and an in-between chamber for soil-sample installation. The active and passive chambers had an inner diameter of 80 mm and a wall thickness of 10 mm, while the chamber for soil-sample installation had an inner diameter of 90 mm. The chambers were hermetically connected using bolts and a sandwiched rubber gasket. A ring-knife soil sampler (a hollow ring with a height of 20 mm, an internal diameter of 79.8 mm, and a wall thickness of 1.8 mm) was adopted. The gap between the outer wall of the ring-knife and the inner wall of the in-between chamber was sealed using two rubber O-rings to prevent gas flow. A perforated circular plate (90 mm in diameter and 5 mm in thickness) with uniformly distributed holes (4 mm in diameter) was installed at both sides of a soil sample under measurement to prevent soil collapse during measurement. Nitrogen (N_2) was supplied to the passive chamber from a vessel connected to the passive chamber, with a pressure control valve and a gas flow control valve inserted in-line to

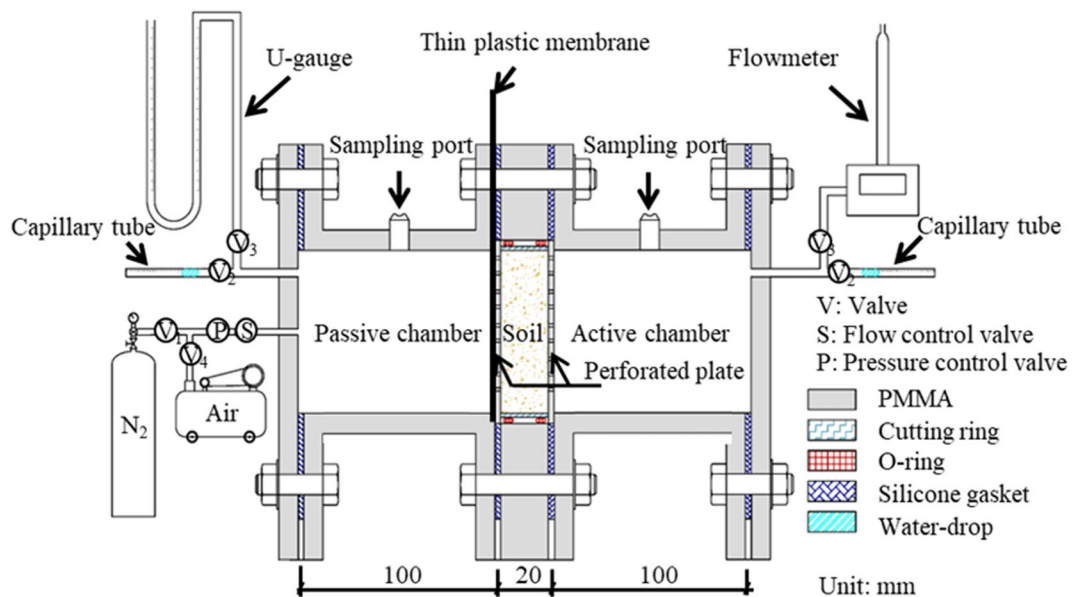


Fig. 1 Schematic of a unified apparatus for measuring gas permeability and gas diffusion coefficient of unsaturated soil (after Feng et al. 2022b)

maintain a constant rate of influx. Capillary tubes (with an inner diameter of 4 mm) containing a water droplet were installed in the active and passive chambers to maintain the atmospheric pressure in both chambers, thereby eliminating the influence of gas advection caused by gas pressure differences during D_p measurement. The volume of the capillary tube was about 13 mL, accounting for only 1.25% of the total volume of the active and passive chambers. Therefore, effects of capillary tube on the gas concentration in the active and passive chambers were expected to be negligible. A gas flow meter with an accuracy of ± 0.1 mL/min was connected to the active chamber to measure the gas flux through the soil sample. A U-gauge with an accuracy of ± 0.01 kPa was connected to the passive chamber to measure gas pressure. 2-mL gas sample was taken through gas sampling ports in the active and passive chambers using a 5 mL syringe. Then, each gas sample was analysed using a gas chromatograph equipped with a thermal conductivity detector (with an accuracy of $\pm 0.1\%$ volumetric concentration of gas) to determine the concentrations of N_2 and O_2 . Before each test, the air tightness was measured by immersing the apparatus in water with all valves closed (V1–V4) and then subjecting it to a gas pressure of 10 kPa, which was greater than the maximum gas pressure (about 5 kPa) applied during tests. No leak was detected, suggesting that the apparatus was air tight during tests. The measurements of k_a and D_p were conducted in a lab with a controlled temperature of 20 ± 1 °C.

Measurement of the D_p of unsaturated soil

After soil sample installation, valves V1 and V2 (Fig. 1) connected to the passive chamber were opened, while valves V3 and V4 were closed. Then, a 0.1 mm-thick plastic membrane was installed between the passive chamber and the soil sample installation chamber, thereby preventing gas exchange between these chambers via the soil before D_p measurement. Pure N_2 was injected into the passive chamber until the measured O_2 concentration in the passive chamber was negligible ($< 0.1\%$). Next, valve V1 was closed to stop N_2 influx, and a water drop was simultaneously added into each capillary tube. The initial O_2 concentration in the active chamber was measured via gas chromatography. Subsequently, the thin plastic membrane sandwiched between the passive chamber and the soil sample was

removed, which allowed gas to diffuse through the soil. Thereafter, the O_2 concentration in the active chamber was measured every 10–15 min (Berisso et al. 2013). According to Glauz and Rolston (1989), the relationship between the D_p of soil (m^2/s) and O_2 concentration in the active chamber is expressed as follows:

$$D_p = - \frac{\ln \left\{ \frac{B(\alpha_1)}{A(\alpha_1)} \left[\frac{1}{2(1+L\epsilon/H)} - \frac{C_i}{C_0} \right] \right\} L^2 \epsilon}{\alpha_1^2 t} \quad (1)$$

where t is the measurement duration (s), L is half the soil-sample height (10 mm), H is the height of the active diffusion chamber (100 mm), C_0 is the initial volumetric concentration of O_2 (m^3/m^3) in the active chamber before the test, and C_i is the O_2 concentration in the active chamber after t (m^3/m^3). When the sizes of the active and passive chambers are the same, $A(\alpha_1)$ and $B(\alpha_1)$ in Eq. (1) are determined as follows (Reible and Shair 1982):

$$A(\alpha_1) = -\frac{1}{\beta^2} - \alpha_1^2 \quad (2)$$

$$B(\alpha_1) = \alpha_1^4 \beta + \alpha_1^2 \left(\frac{2}{\beta} + 1 \right) + \frac{1}{\beta^3} + \frac{1}{\beta^2} \quad (3)$$

$$\alpha_1^2 = \frac{1}{\beta} - \frac{1}{3\beta^2} + \frac{4}{45\beta^3} + \frac{16}{945\beta^4} \quad (4)$$

$$\beta = \frac{H}{L\epsilon} \quad (5)$$

Thus, to determine D_p , the measured time series of O_2 concentration (C_i) was best-fitted using Eq. (1). During D_p measurement (which lasted for ~45 min), the soil respiration-induced change in O_2 concentration was ~0.1%, which was considerably lower than that observed in the active chamber ($> 5\%$). Therefore, the effect of soil respiration on O_2 concentration was ignored in this study.

Measurement of the k_a of unsaturated soil

k_a was measured as follows. First, valves V1 and V2 (Fig. 1) were closed, while valves V3 and V4 were opened. Then, air was pumped into the passive chamber from the air compressor at a constant flow rate.

Approximately 1–5 min after the start of pumping, the U-gauge reading was stable, so a reading was taken. The air flow rate of the active chamber was measured using a flowmeter, and then the aforementioned test procedure was repeated. The air flow rate was regulated using a flow control valve, so that the gas pressures in the passive chamber could be obtained under at least three air flow rates. According to Darcy's law, the k_a of soil can be calculated as follows (Bouazza and Vangpaisal 2003):

$$k_a = \frac{Q\mu}{A} \frac{x}{\Delta P} \quad (6)$$

where Q is the air flow rate out of the active chamber (m^3/s), μ is the dynamic viscosity of the gas ($\text{Pa}\cdot\text{s}$), A is the cross-sectional area of the soil sample (0.005 m^2), ΔP is the gas pressure measured in the passive chamber (Pa), and x is the soil sample height (0.02 m). The measured relationship between the gas pressure gradient ($\frac{\Delta P}{x}$) and the gas flow velocity (i.e., Darcian velocity $v = \frac{Q}{A}$; m/s) was best-fitted using Eq. (6) to obtain k_a .

Because Darcy's law is only valid for the laminar flow of fluid in porous media, controlling the Reynolds number of air flow (R_e) is necessary. R_e is defined in Eq. (7), as follows:

$$R_e = \frac{vd}{\gamma} \quad (7)$$

where d is the average particle diameter of the pore matrix ($d_{50} = 7 \times 10^{-4} \text{ m}$, Table 1), and γ is the kinematic viscosity of fluid ($1.48 \times 10^{-5} \text{ m}^2/\text{s}$ for air at 20°C and an atmospheric pressure of 101.3 kPa). At $R_e < 1$, the laminar flow assumption and hence Darcy's law are valid (Bouazza and Vangpaisal 2003; Rouf et al. 2016). Accordingly, the maximum air flow rate was controlled at $7.5 \times 10^{-6} \text{ m}^3/\text{s}$ (i.e., 0.45 L/min), corresponding to a maximum R_e of 0.07 (see details in Table S1).

Analysis of soil microstructure

After the k_a and D_p measurements of soil samples with degrees of compaction of 80% and 95%, which represented the lowest and highest considered degrees of compaction, cubic soil specimens of $\sim 0.5 \text{ cm}^3$ were collected for soil microstructure analysis (Chen et al. 2020). Prior to analysis of soil microstructure, the

cubic soil specimens were dried by freeze drying. The soil microstructure was observed via scanning electron microscopy (Quanta 250, FEI Company, USA; Zhan et al. 2022), including pore-size and pore shape of soil matrix, and the interface between soil matrix and root. The observations were made using a Quanta 250 scanning electron microscope (FEI Company, USA). Soil pore-size distributions were quantified via mercury intrusion porosimetry following the standard test procedures as outlined in ASTM (2018). An AutoPore IV 9500 porosimeter (Micromeritics Instruments Co., Ltd., USA) was used. The porosimeter determines the pore size distribution of soil based on the principle that the mercury intrusion pressure corresponds to the pore size following Young–Laplace's law (Nguyen et al. 2021). The pore size range determined herein was between 3 nm and $400 \mu\text{m}$.

Measurement of root characteristics

The sample soil particles were carefully washed with water to extract the roots. The root characteristics—the root area index (a ratio of the total root surface area to the plane cross-sectional area of soil for a given root depth, Ng et al. 2016); the root area ratio (the ratio of the total cross-sectional area of all roots in a horizontal cross section at a certain depth to the total cross-sectional area of soil at that depth, Ng 2017), R_v , and root length density—were quantified via image analysis using WinRHIZO software, which measures root characteristic parameters from images according to the skeletonisation method principle (Arsenault et al. 1995). Pang et al. (2011) found that the root volume measured by WinRHIZO software was close to that measured via the immersion method, with a maximum error less than 1%. In the present study, root mass was measured using an electronic scale with an accuracy of 0.001 g , which enabled the root biomass ratio (R_b , the dry mass of roots per unit dry soil mass) to be obtained. The measured root characteristic parameters are presented in Table 2.

Grey relational analysis

The relative effects of various root characteristic parameters on the k_a and D_p of rooted soil were compared through grey relational analysis. Grey relational analysis is an effective method for quantifying and comparing

Table 2 Root characteristic parameters of soil samples

Specimen ID	Root area index	R_v	Root area ratio (10^{-5})	R_b (10^{-4} g/cm ³)	Root length density (cm/cm ³)
80–40–1*	3.984	0.010	15.0	6.5	1.554
80–40–2	7.807	0.027	39.9	7.6	2.538
80–40–3	5.333	0.019	27.6	5.2	1.617
80–60–1	3.229	0.010	11.3	5.0	1.474
80–60–2	2.837	0.007	9.5	2.7	1.223
80–60–3	3.612	0.011	27.2	4.1	1.198
80–80–1	1.436	0.005	8.2	3.8	0.631
80–80–2	1.764	0.006	8.7	2.8	0.802
80–80–3	2.259	0.007	4.3	2.3	1.234
80–100–1	2.505	0.005	4.9	3.8	1.210
80–100–2	0.725	0.002	4.0	2.3	0.280
80–100–3	1.211	0.004	3.8	1.8	0.399
85–40–1	15.044	0.034	17.9	5.9	6.795
85–40–2	23.346	0.051	27.6	16.2	11.013
85–40–3	21.859	0.046	33.9	7.9	10.464
85–60–1	10.980	0.016	8.6	2.9	1.636
85–60–2	14.383	0.032	11.8	4.3	6.523
85–60–3	12.731	0.028	14.1	4.0	2.734
85–80–1	6.250	0.013	1.8	2.2	3.104
85–80–2	8.303	0.018	3.5	1.9	3.858
85–80–3	13.824	0.027	9.9	3.4	7.122
85–100–1	7.938	0.017	2.7	2.3	3.639
85–100–2	12.044	0.024	3.7	2.2	6.211
85–100–3	6.832	0.013	3.5	0.9	3.658
90–40–1	3.855	0.008	14.7	14.1	1.859
90–40–2	9.124	0.022	26.9	8.8	3.767
90–40–3	7.478	0.018	24.6	5.1	3.146
90–60–1	3.855	0.008	12.6	3.6	1.485
90–60–2	6.049	0.017	15.5	5.2	2.122
90–60–3	3.055	0.008	10.1	4.4	1.231
90–80–1	3.490	0.009	14.5	2.5	1.369
90–80–2	1.687	0.004	2.6	2.2	0.750
90–80–3	1.208	0.010	10.5	3.8	1.723
90–100–1	2.170	0.005	6.3	1.9	0.977
90–100–2	1.259	0.003	3.9	2.2	0.455
90–100–3	1.359	0.003	2.5	1.5	0.668
95–40–1	14.728	0.030	18.6	8.0	7.221
95–40–2	12.113	0.026	19.2	8.3	5.720
95–40–3	10.901	0.022	17.1	9.2	2.895
95–60–1	8.948	0.019	11.5	4.4	4.339
95–60–2	9.342	0.016	8.2	8.4	3.416
95–60–3	7.208	0.014	4.0	2.4	3.678
95–80–1	10.417	0.021	7.5	9.1	5.201
95–80–2	4.232	0.009	4.5	0.6	1.901
95–80–3	7.563	0.014	7.9	2.1	1.505
95–100–1	2.641	0.006	6.6	2.1	1.099
95–100–2	3.852	0.007	2.6	1.0	2.056
95–100–3	1.164	0.002	0.8	1.9	0.563

*Meaning of specimen ID: ①–②–③: ① represents the degree of compaction of the soil (e.g., 80 represents 80% degree of compaction); ② represents the depth where the soil was taken (e.g., 40 denotes the depth of 40 mm); ③ represents the numeration of the soil sample taken under a certain degree of compaction and soil depth (e.g., 1 represents sample No. 1)

the effects of various factors (i.e., root characteristics) on measurements (i.e., k_a and D_p). The advantages of grey relational analysis include its requiring only limited data on the relationship between multiple factors in a complex system (Xu et al. 2018). The correlations between the influencing factors and k_a (or D_p) were quantified using a correlation coefficient that could take a value from 0 to 1, where a large value shows that the related factor has a large influence. In the present study, the coefficient of correlation between each root characteristic parameter and k_a or D_p was calculated using MATLAB (see Code S1) according to the measured k_a , D_p , and root characteristics. It is known that k_a and D_p were not only affected by root characteristics, but also by gas-filled porosity ε . To eliminate effect of ε on the k_a and D_p of soil, datasets with similar ε values (i.e., with a maximum relative difference < 5%) were selected for grey relational analysis (Table S2). Thus, any differences of k_a or D_p within the same dataset of similar ε were primarily due to differences in root characteristics.

Empirical predictive models of the k_a and D_p of rooted soil

Existing models for predicting the k_a of bare soil typically consider the influences of ε and pore connectivity (Ball et al. 1988; Chen et al. 2014; Hamamoto et al. 2009, 2011; among many others). The general form of the most widely used models for predicting the k_a of bare soil is expressed as follows (Hamamoto et al. 2011):

$$k_a = \alpha_a \cdot \varepsilon^{X_a} \quad (8)$$

where α_a is a constant reflecting the connectivity and tortuosity of soil pores (m^2) and X_a is a dimensionless constant depicting the effects of water blockage on gas movement (Chen et al. 2017; Zhan et al. 2014). For rooted soil, α_a and X_a are replaced by α_r and X_r , respectively, which are considered functions of root characteristic parameters (e.g., R_v) and defined as follows:

$$\alpha_r = f(R_v) \quad (9)$$

$$X_r = F(R_v) \quad (10)$$

where $f(R_v)$ and $F(R_v)$ were obtained from rooted soil measurements (as detailed in Eqs. (14) and (15) given later). R_v is used in Eqs. (9) and (10) because it has a greater influence on k_a and D_p than the other selected root characteristic parameters (as detailed in Section 5.1).

Various models have been devised for predicting the D_p of bare soil (Hamamoto et al. 2011, 2012; Moldrup et al. 2000, among many others) and they generally have similar exponential forms, such as (Hamamoto et al. 2012):

$$\frac{D_p}{D_0} = (\varepsilon - \varepsilon_{th})^{X'} \quad (11)$$

where D_0 is the D_p in free air ($2.0 \times 10^{-5} \text{ m}^2/\text{s}$ for O_2 at 20°C ; Hamamoto et al. 2012); ε_{th} is the percolation threshold, below which gas cannot diffuse through the gas phase in the soil (i.e., $D_p=0$); and X' is an empirical factor that considers pore tortuosity, which is affected by soil texture, structure, and moisture content. Generally, at a small ε (e.g., at an ε less than that corresponding to the saturation degree of 85%), the gas phase in soil becomes discontinuous, so gas can only diffuse through the water phase (Zhan et al. 2016). Resurreccion et al. (2008) found that under such conditions, gas diffusion was negligible because the diffusion coefficient of gas in the water phase was approximately four orders of magnitude lower than that in the gas phase. Schjønning et al. (2003) suggested that the percolation threshold (ε_{th}) in Eq. (11) could be taken as the ε corresponding to a relative gas diffusion coefficient (D_p/D_0) of 1×10^{-4} , which is calculated using Eq. (11) under the assumption that ε_{th} is 0.

For rooted soil, $\varepsilon_{th,r}$ and X'_r are taken as functions of root characteristic parameters (such as R_v), as follows:

$$\varepsilon_{th,r} = g(R_v) \quad (12)$$

$$X'_r = G(R_v) \quad (13)$$

where $g(R_v)$ and $G(R_v)$ are obtained using the test data of the rooted soil (as detailed in Sect. 5.5). Thus, the D_p of rooted soil was estimated using Eqs. (11), (12), and (13).

Results

Coefficient of correlation between root characteristics and k_a or D_p

Figure 2a shows typical measured and fitted results used for determining the k_a of rooted soils with different gas contents. The figure reveals the linear relationships between the gas flow velocity (v) and the gas pressure gradient (dP/dx). The measurements were best-fitted using Eq. (6) to determine k_a , and the goodness of fit (i.e., coefficient of determination (R^2)) was greater than 95%. Figure 2b shows the time series of measured O_2 concentrations in the active chamber. The D_p was determined through curve-fitting using Eq. (1) and its R^2 was also greater than 95%. These results demonstrate that the k_a and D_p of the soil samples was effectively measured using Eqs. (6) and (1), respectively.

Through grey relational analysis, the coefficients of correlation between each root characteristic parameter and the k_a (or D_p) of soil samples with different degrees of compaction and ϵ values were obtained (Table 3). All the correlation coefficients differ significantly from zero (p -value < 0.05). Among the selected parameters, R_v generally exerted the greatest influence on the k_a and D_p of soil, possibly because roots altered soil pore size and hence affected the tortuosity and available pore space for gas movement. Overall, the correlation coefficient of R_b was lower than that of R_v , indicating that the soil samples differed in root biomass density; otherwise, R_b and R_v would have been interconvertible by using a constant root biomass density (i.e., R_b and R_v would have been equivalent).

k_a of rooted soil

Figure 3a shows that at a degree of compaction of 80%, the k_a of the rooted soil was similar with that of the bare soil. At higher degree of compaction – i.e. 85% (Fig. 3b) and 90% (Fig. 3c) – the rooted soil generally exhibited a larger k_a than the bare soil. Moreover, the difference was more significant in soil with a high degree of compaction and a large ϵ . In addition, the rooted soil with a degree

of compaction of 95% exhibited a much higher k_a than the bare soil in the entire ϵ range considered (Fig. 3d), possibly because the roots along the root–soil interface increased the gas preferential flow through larger pores (as detailed below in this section). These results indicate that the effects of roots on k_a were more significant in soil with a higher degree of compaction and with higher values of ϵ than in soil with a lower degree of compaction and with lower values of ϵ . Figure 3c shows that at $R_v < 0.005$, the rooted soil generally exhibited substantially higher k_a than the bare soil. At $R_v > 0.005$, the k_a of the rooted soil generally decreased with increasing R_v for a given ϵ , possibly because root occupation reduced soil pores, resulting in a smaller soil pore size.

The effects of roots on the k_a of soil could be explained by the soil microstructure. The bare and rooted soils with 80% degree of compaction exhibited relatively large macropores owing to their low degree of compaction (Fig. 4a and b). In the current study, although arc-shaped macropores (i.e., the large void space between roots and soil matrix; Fig. 4b) existed along the root–soil interface, their sizes were similar to those of the macropores in the bare soil (Fig. 4a). Consistently, measurements of mercury intrusion porosimetry (Fig. 5) showed that although the rooted soil contained fewer non-occupied macropores than the bare soil at 80% degree of compaction, the rooted soil nevertheless exhibited relatively large macropores ($\sim 3 \times 10^4$ nm); therefore, the macropores were the dominant factor influencing k_a (Berisso et al. 2013). Thus, the existence of arc-shaped macropores along the root–soil interface did not significantly affect the k_a of soil with an 80% degree of compaction (Fig. 3a). The rooted and bare soils with a 95% degree of compaction (Fig. 4c and d) exhibited much smaller pores than those with an 80% degree of compaction. Nevertheless, at a degree of compaction of 95%, the arc-shaped macropores (Fig. 4d) along the root–soil interface were much larger than the pores in the bare soil (Fig. 4c). This was substantiated by the measured pore size distributions (Fig. 5), which showed that at a 95% degree of compaction, the rooted soil contained more macropores than the bare soil.

D_p of rooted soil

Figure 6 depicts the relationships between D_p and ε for rooted soils with different R_v and degree of compaction values. The bare and rooted soils with an 80% degree of compaction had similar D_p values, possibly because the sizes of the pores around their root–soil interfaces were comparable to those of the bare soil (Fig. 4a and b) owing to the low degree of compaction. Regardless of the degree of compaction and R_v , the rooted and bare soils exhibited comparable D_p values at $\varepsilon < 0.1$. As ε exceeded 0.10, roots generally increased the D_p of rooted soil with a degree of compaction $\geq 85\%$ by a factor of several units (Fig. 6b–d), which is attributable to the improved pore connectivity resulting from the root-induced arc-shaped macropores along the root–soil interface.

Effects of R_v on k_a and D_p

Figure S1 shows the relation between R_v and k_a of soils with different degrees of compaction. The data are scattered and lack a consistent pattern in soils with a degree of compaction of 80% (Fig. S1a) and 85% (Fig. S1b), respectively. Under these two relatively low degrees of compaction, the soil pore size was comparable to the macropore size along the root–soil interface (Fig. 4a and b), which limited

the influence of the roots on k_a . At a high degree of compaction ($\geq 90\%$), k_a tended to decrease with increasing R_v (Figs. S1c and d; R^2 is 0.66 and 0.45 for Figs. S1c and d, respectively), suggesting that the effect of root occupancy in soil pores was greater than the effect of root-induced macropores under these conditions, because (i) more pores were filled by more roots and (ii) the size of the macropores along the root–soil interface reduced as roots coarsened as they grew (Zhan et al. 2016). Regardless of the degree of compaction, the k_a generally decreased as ε decreased for a given R_v ; this indicated that compared with R_v , ε affected k_a more significantly.

Figure S2 depicts the relationship between the R_v and D_p of the rooted soil at $\varepsilon > 0.10$. At $\varepsilon < 0.10$, the high-water content dominated D_p (Fig. 6) owing to the reduced gas phase connectivity. Similar to the influence of R_v on k_a (Fig. S1), a larger R_v generally reduced D_p for a given ε interval at a degree of compaction $\geq 90\%$ (Figs. S2c and d), because roots filled the soil pores under these conditions. Generally, roots had a greater effect on k_a than on D_p . Specifically, k_a was reduced by approximately one order of magnitude as R_v increased from 0.002 to 0.02 at an ε range of 0.10–0.12 (Fig. S1c), while D_p was only decreased by half (Fig. S2c).

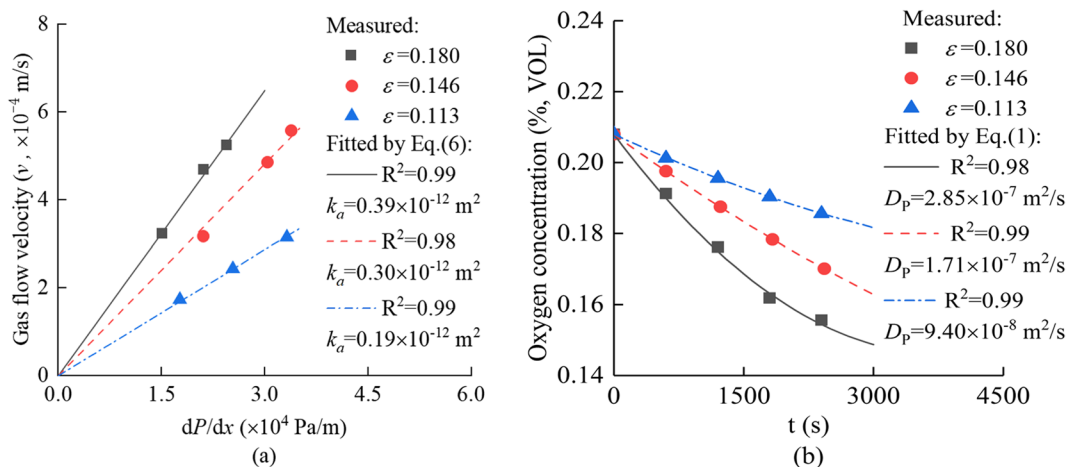


Fig. 2 Typical measured and fitted results for determining (a) gas permeability and (b) gas diffusion coefficient of rooted soil with a degree of compaction of 90% and R_v of 0.018

Empirical predictive models of the k_a and D_p of rooted soil

An empirical predictive model was established through the best-fitting of the measured k_a – ε relationship (Fig. 3) using Eqs. (8)–(10). Accordingly, α_r (Eq. 9) and X_r (Eq. 10) were established, as follows:

$$\alpha_r = f(R_v) = \begin{cases} \alpha_a + 3.66 \times 10^{-8} R_v - 1.47 \times 10^{-10} & (80\% \text{ DOC}) \\ \alpha_a + 1.83 \times 10^{-9} R_v + 7.83 \times 10^{-12} & (85\% \text{ DOC}) \\ \alpha_a + 5.63 \times 10^{-7} R_v^2 + 1.05 \times 10^{-8} R_v + 2.83 \times 10^{-12} & (90\% \text{ DOC}) \\ \alpha_a + 3.38 \times 10^{-9} R_v + 8.49 \times 10^{-11} & (95\% \text{ DOC}) \end{cases} \quad (14)$$

$$X_r = F(R_v) = \begin{cases} X_a + 7.23 R_v - 0.03 & (80\% \text{ DOC}) \\ X_a + 4.78 R_v - 0.22 & (85\% \text{ DOC}) \\ X_a - 6435.90 R_v^2 + 169.44 R_v - 0.54 & (90\% \text{ DOC}) \\ X_a - 31.32 R_v + 0.68 & (95\% \text{ DOC}) \end{cases} \quad (15)$$

where α_a and X_a (see Table S3) were determined through the best-fitting of the measurements of the bare soil k_a at each degree of compaction using Eq. (8), which reasonably depicts the k_a of the bare soil (see Fig. S3). However, the predictive model of the bare soil either overestimated (Fig. 7a) or underestimated (Fig. 7d) the measured k_a by up to more than one magnitude, because the models do not

consider the soil pores previously occupied by roots. The predictive models of the rooted (Eqs. 14 and 15) and bare soils with an 80% degree of compaction gave similar results (Fig. 7a). In contrast, the predictive models of the rooted soil with a degree of compaction $\geq 85\%$ yielded results that are more consistent with the measurements (Fig. 7b to d). These results demonstrate the importance of considering the effects of roots on k_a when studying dense soil (i.e., soil with a degree of compaction $\geq 85\%$), which is commonly used in infrastructures (e.g., landfill cover and man-made slopes; Zhan et al. 2020) to ensure slope stability. Conversely, the effects of roots of Bermuda grass on the k_a of soil might be ignored for the relatively loose silty sand with a degree of compaction of 80% studied herein.

The $\varepsilon_{th,r}$ (Eq. 12) and X_r (Eq. 13) of the rooted soil with different degrees of compaction were obtained through the best-fitting of the measured D_p (see Fig. 6) using Eq. (11). Figure 8 compares the percolation thresholds of the rooted ($\varepsilon_{th,r}$) and bare soils (ε_{th}). Rooted soils generally exhibited lower percolation thresholds than bare soils under various degrees of compaction. For the rooted soil, its lower threshold of gas diffusion is attributable to the pores around the root–soil interface being larger than the pores in the bare soil; this is

Table 3 Correlation coefficient of root characteristic parameters with gas permeability or gas diffusion coefficient

Degree of compaction (%)	ε (m ³ /m ³)	Correlation coefficient of root characteristic parameters with k_a or D_p									
		Root area index (-) ^a		Root volume ratio (-)		Root area ratio (-)		Root biomass ratio (g/cm ³)		Root length density (cm/cm ³)	
		k_a	D_p/D_0	k_a	D_p/D_0	k_a	D_p/D_0	k_a	D_p/D_0	k_a	D_p/D_0
80	0.24	0.67	0.62	0.82	0.77	0.86	0.83	0.83	0.79	0.67	0.60
	0.17	0.52	0.62	0.82	0.88	0.69	0.78	0.60	0.68	0.63	0.71
	0.11	0.79	0.78	0.84	0.83	0.62	0.61	0.59	0.58	0.76	0.74
85	0.16	0.78	0.76	0.97	0.97	0.82	0.81	0.64	0.63	0.70	0.69
	0.12	0.83	0.82	0.86	0.85	0.81	0.79	0.80	0.79	0.71	0.69
	0.10	0.91	0.86	0.99	0.98	0.86	0.80	0.88	0.83	0.83	0.87
90	0.18	0.94	0.92	0.96	0.95	0.86	0.84	0.68	0.64	0.77	0.74
	0.15	0.93	0.92	0.96	0.95	0.93	0.91	0.67	0.61	0.74	0.70
	0.12	0.92	0.90	0.93	0.91	0.84	0.81	0.58	0.53	0.60	0.57
95	0.12	0.79	0.72	0.97	0.95	0.81	0.73	0.79	0.72	0.78	0.70
	0.1	0.83	0.81	0.96	0.95	0.73	0.71	0.72	0.70	0.62	0.59
	0.07	0.83	0.85	0.89	0.90	0.75	0.77	0.45	0.49	0.57	0.61
Mean correlation degree		0.81	0.80	0.91	0.91	0.80	0.78	0.69	0.67	0.70	0.68

a: "-" represents dimensionless

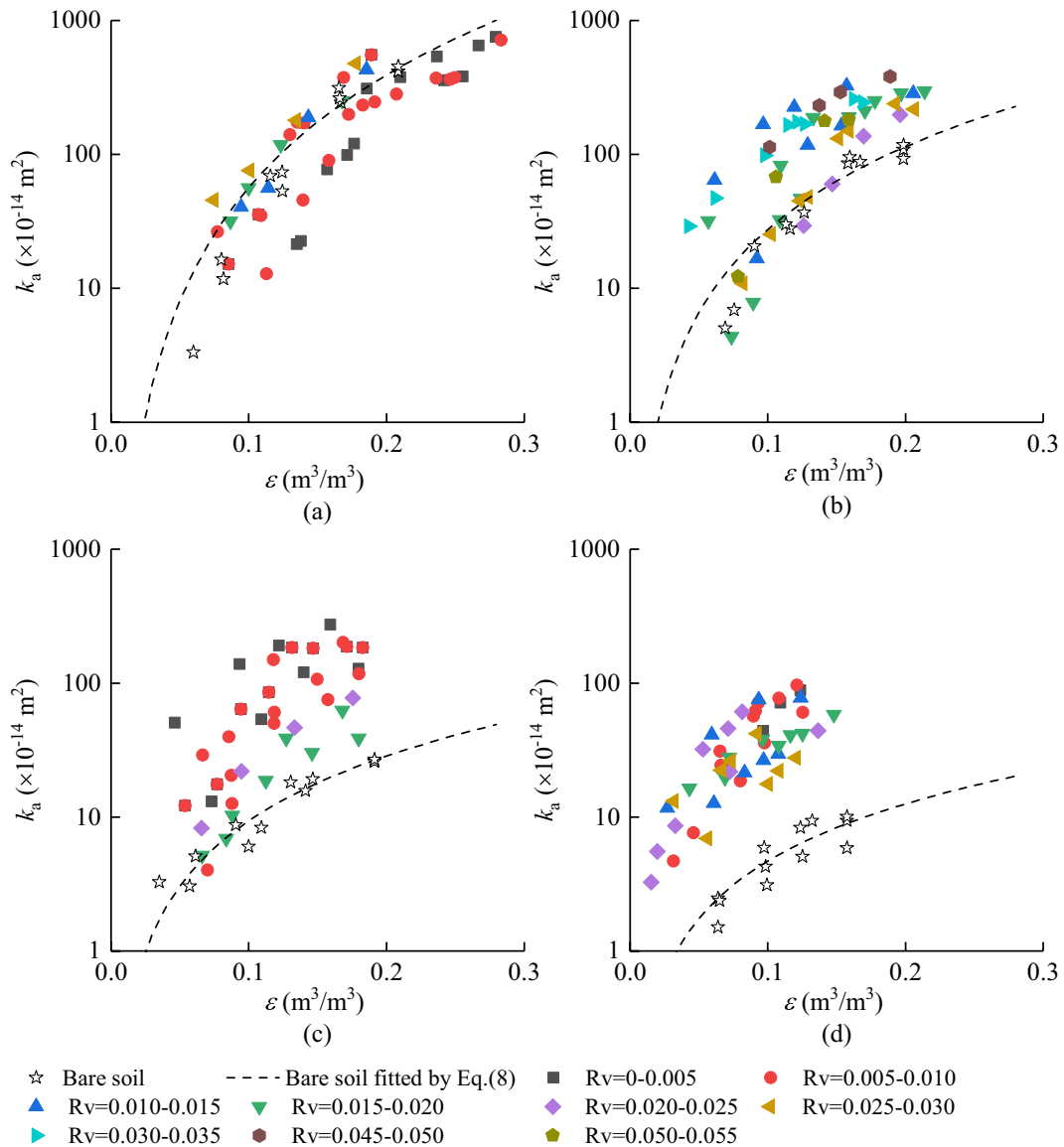


Fig. 3 Relationship between gas permeability and gas content of rooted soil with different degrees of compaction: **a** 80% degree of compaction; **b** 85% degree of compaction;

c 90% degree of compaction; **d** 95% degree of compaction (Data point and dashed line represent measurement and fitting results, respectively.)

because larger pores increase gas phase connectivity in soil, as macropores are less prone to water blockage than micropores. Accordingly, the $\varepsilon_{th,r}$ and X'_r of the rooted soil were empirically obtained, as follows:

$$\varepsilon_{th,r} = g(R_v) = \begin{cases} \varepsilon_{th} - 0.312R_v - 0.007 & (80\% \text{ DOC}) \\ \varepsilon_{th} - 0.080R_v - 0.008 & (85\% \text{ DOC}) \\ \varepsilon_{th} + 0.350R_v - 0.017 & (90\% \text{ DOC}) \\ \varepsilon_{th} - 0.080R_v - 0.056 & (95\% \text{ DOC}) \end{cases} \quad (16)$$

$$X'_r = G(R_v) = \begin{cases} X' - 7.54R_v - 0.078 & (80\% \text{ DOC}) \\ X' - 2.12R_v - 0.226 & (85\% \text{ DOC}) \\ X' + 13.58R_v - 0.467 & (90\% \text{ DOC}) \\ X' - 2.243R_v - 0.149 & (95\% \text{ DOC}) \end{cases} \quad (17)$$

where ε_{th} and X' are fitting parameters (Table S4) obtained through the best-fitting of the D_p measurements of the bare soil using Eq. (11), which reasonably depicts the D_p of the bare soil (Fig. S2).

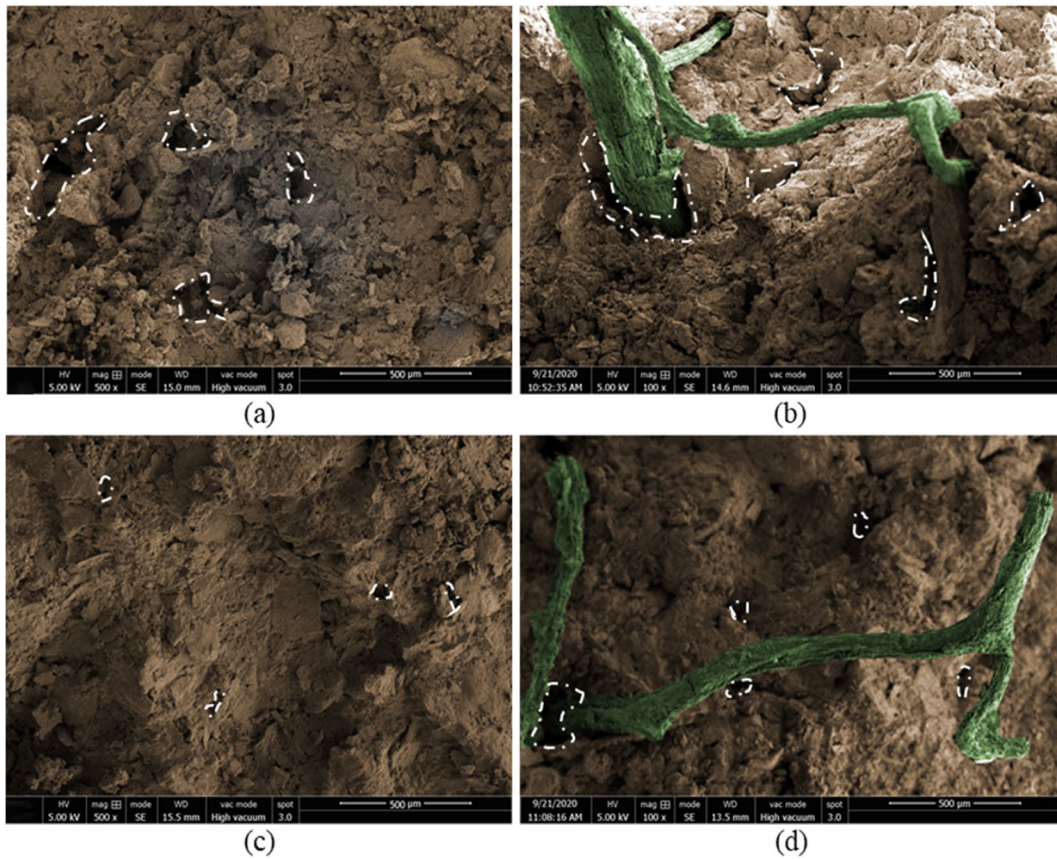
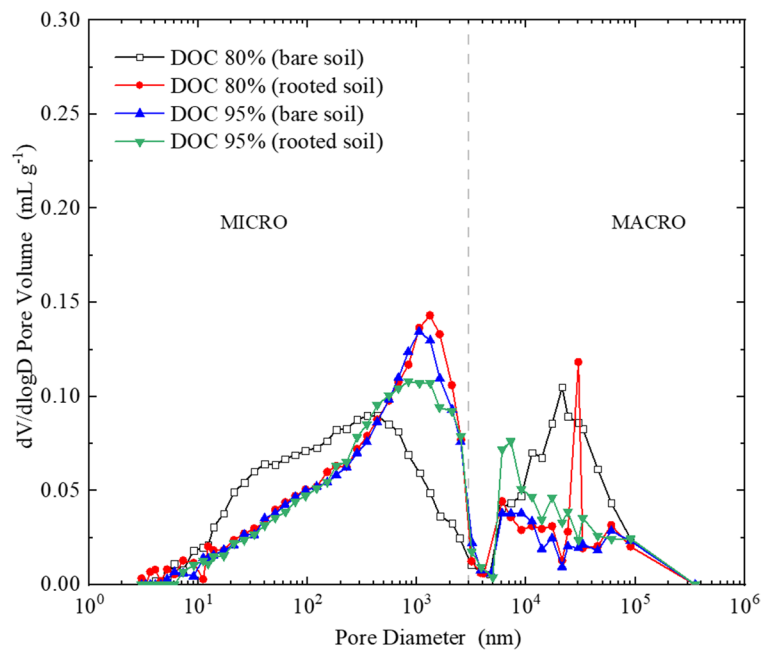


Fig. 4 Microscopic characteristics of different soil samples: **a** bare soil with 80% degree of compaction; **b** rooted soil with 80% degree of compaction; **c** bare soil with 95% degree of compaction; and **d** rooted soil with 95% degree of compaction

Fig. 5 Pore-size distributions for bare and rooted soils with 80% degree of compaction and 95% degree of compaction



The empirical predictive model of D_p considering the effects of R_v was obtained by substituting Eqs. (16) and (17) into the D_p predictive model (Eq. 11). Overall, independent of degree of compaction, the predicted results considering the effects of roots are more consistent with the measurements (Fig. 9), as evidenced by these predicted results having a lower root-mean-square error than those for the bare soil. For an example, the root-mean-square error for rooted

soil and bare soil with a degree of compaction of 90% is 9.6×10^{-3} and 1.2×10^{-2} , respectively. Moreover, for the cases considering and ignoring the effects of roots, respectively, the difference in the predicted D_p (Fig. 9) values was less than the difference in the predicted k_a values (Fig. 7). This suggests that roots have less of an influence on D_p than on k_a . Thus, the empirical predictive model can be used to preliminarily evaluate how gas movement is affected by roots,

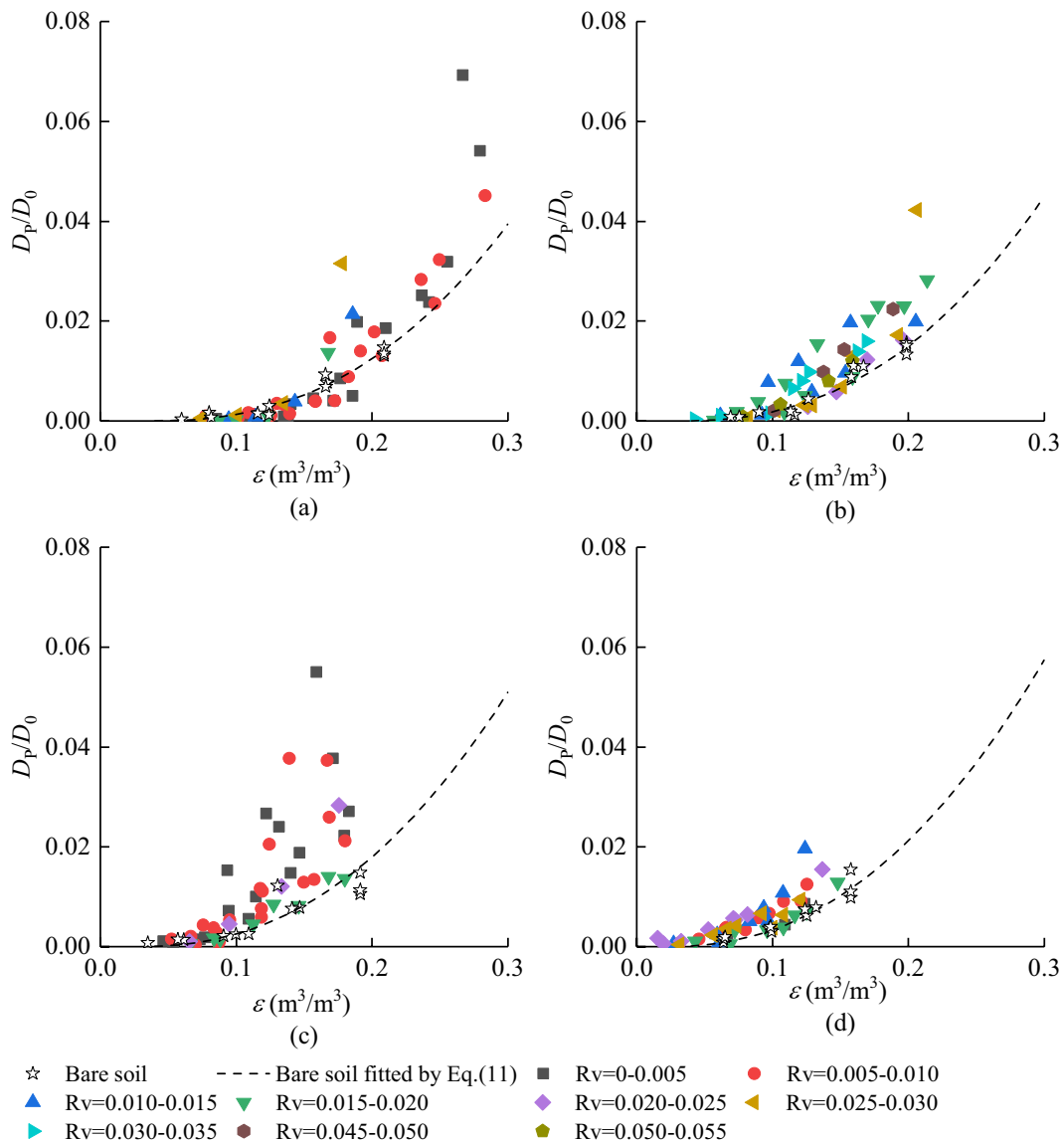


Fig. 6 Relationship between gas diffusion coefficient and soil gas content of rooted soil with different degrees of compaction: **a** 80% degree of compaction; **b** 85% degree of compaction; **c**

90% degree of compaction and **d** 95% degree of compaction (Dashed line and data point represent fitting result by Eq. (11) and measurement, respectively.)

which could be used in final cover preliminary design to reduce landfill gas emissions and O_2 entry. However, the predictive models for both the k_a (Eqs. 14 and 15) and D_p (Eqs. 16 and 17) of the rooted soil were empirically established according to the measurements of the present study. Thus, the predictive models may be applicable to determining the short-term effects (around six months) of roots of *C. dactylon* on k_a and D_p in silty sand with the degree of compaction between 80 and 95%. Caution should be taken in applying these predictive models to other soil types bearing other plant species.

Discussion

The experimental results show that roots affected k_a (Fig. 3) and D_p (Fig. 6) of soil through root–soil interface (Fig. 4b and d), which changed the soil microstructure (e.g., pore size distributions; Fig. 5). The impact of roots on k_a and D_p depended on the degree of compaction and soil gas content of soil matrix. When the degree of compaction $\geq 90\%$ and soil gas content ≥ 0.10 , roots generally increased k_a (Fig. 3c and d) and D_p (Fig. 6c and d). This section further discusses the mechanisms of root-induced change in soil microstructure on k_a and D_p . Moreover, future researches about effects of roots on k_a and D_p were also discussed.

Both scanning electron microscopy (Fig. 4) and mercury intrusion porosimetry (Fig. 5) results show that roots increased macropores in compacted soil through the arc-shaped macropores along the root–soil interface. Consistently, Carminati et al. (2013) also reported that the gap between roots and soil with a matric potential of -50 kPa reached 0.10 – 0.35 mm. The difference between root-induced macropores along root–soil interface and pores of soil matrix became more noticeable as the degree of compaction increased (Fig. 4). This is consistent with Kautz et al. (2010), who found that roots alleviate compaction. Indeed, bio-tillage has recently been proposed to utilize plant roots to alleviate soil compaction, because roots reduce soil bulk density and increase macro-aggregate (Ali et al. 2022), leading to an increase in the saturated hydraulic conductivity and gas permeability and hence improvement of nutrition supply and soil aeration for crop growth (Zhang et al. 2022; Zhang and Peng 2021). Accordingly, the arc-shaped

macropores along the root–soil interface increased the k_a of soil with a degree of compaction of 95% (Fig. 3d), which was attributable to (i) the much larger gas advective flow velocity, as indicated by the Hagen–Poiseuille equation (i.e., the average advective flow velocity is proportional to the square of pore radius; Berisso et al. 2013; Nguyen et al. 2021; Zhan et al. 2014) and (ii) the presence of macropores increasing the pore connectivity, as macropores are less susceptible than micropores to water filling, in accordance with the law of capillarity (i.e., a tube with a smaller radius features a higher capillary height than a tube with a larger radius; Kiuru et al. 2022; Nguyen et al. 2021). k_a also depended on the tortuosity of the gas flow path, which was likely affected by the heterogeneously spatial distribution of roots. Thus, the k_a of the rooted soil exhibited pronounced scattering even within the same R_v range (Fig. 3c) for a given ε . In contrast, Zhan et al. (2016) found that the growth of grass roots led to an order-of-magnitude decrease in the k_a of compacted loess with a degree of compaction of 85% , owing to the presence of roots in macropores. Moreover, Ni and Ng (2019) reported that at a degree of compaction of 95% , the k_a of rooted soil was lower than that of bare soil after 6-months of grass growth, whereas this trend was reversed after grass grew for over 15 months. These results indicate that the roots had a dual impact on the k_a of soil: (i) they reduced the number and size of pores available for gas movement by occupying soil pores, thereby reducing k_a ; (ii) their decay or shrinkage upon desiccation increased soil pore size along the root–soil interface (Liu et al. 2015), thereby increasing k_a .

The increase in k_a by the root was much greater than that of D_p . For an example, the rooted soil generally exhibited a larger k_a than the bare soil by one to two orders of magnitude for the entire ε range for soil with a degree of compaction $\geq 85\%$ (Fig. 3b–d). On the contrary, roots generally increased the D_p of rooted soil by several times, when ε exceeded 0.10 and the degree of compaction $\geq 85\%$ (Fig. 6b–d). This was likely because k_a was affected not only by ε but also by the root-induced macropores (Figs. S1b and d). Specifically, the average advective flow velocity is proportional to the square of pore radius, according to the Hagen–Poiseuille equation (Nguyen et al. 2021; Zhan et al. 2014). Indeed, studies have reported that the k_a of clayey soil increased as its water content

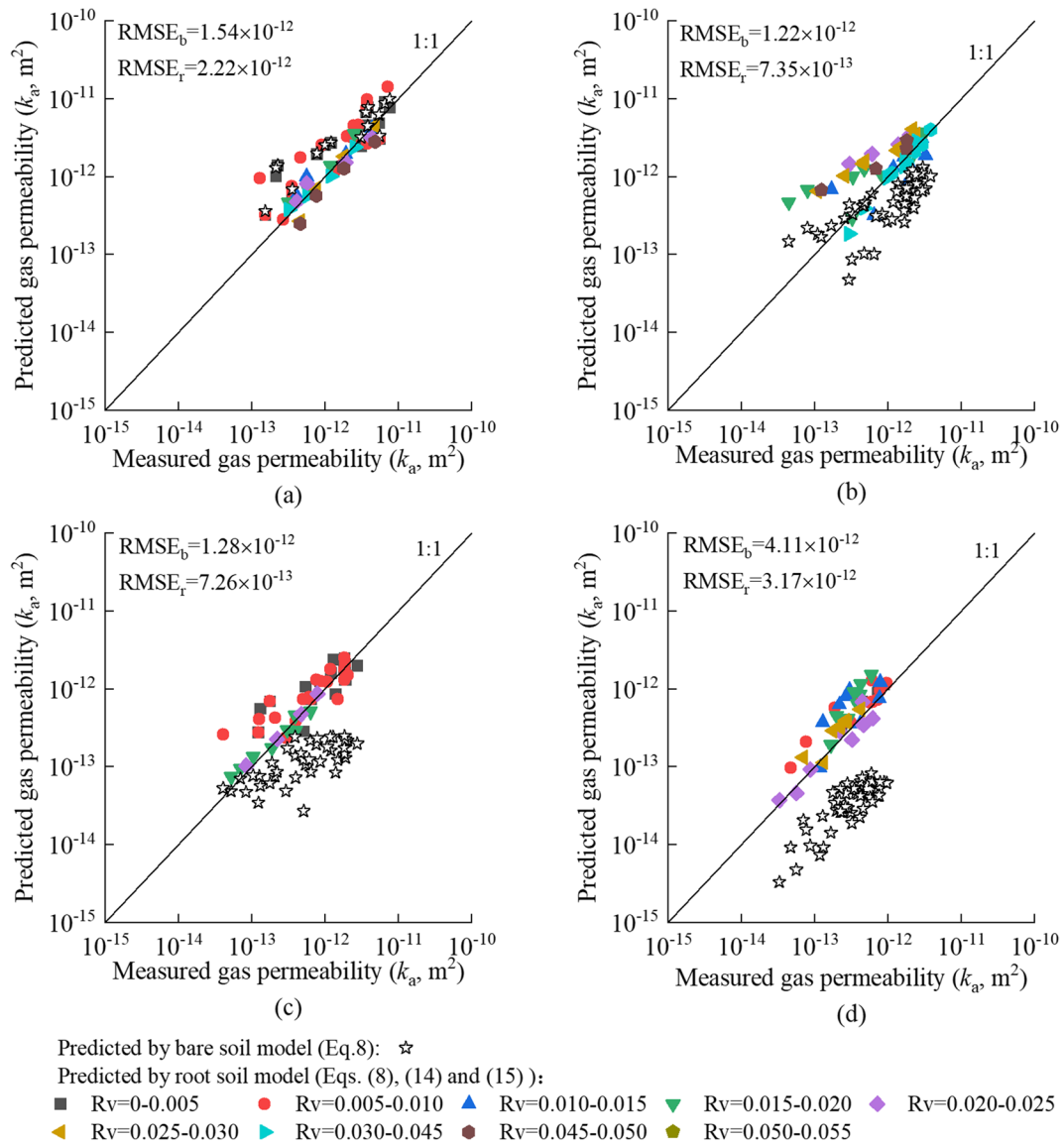
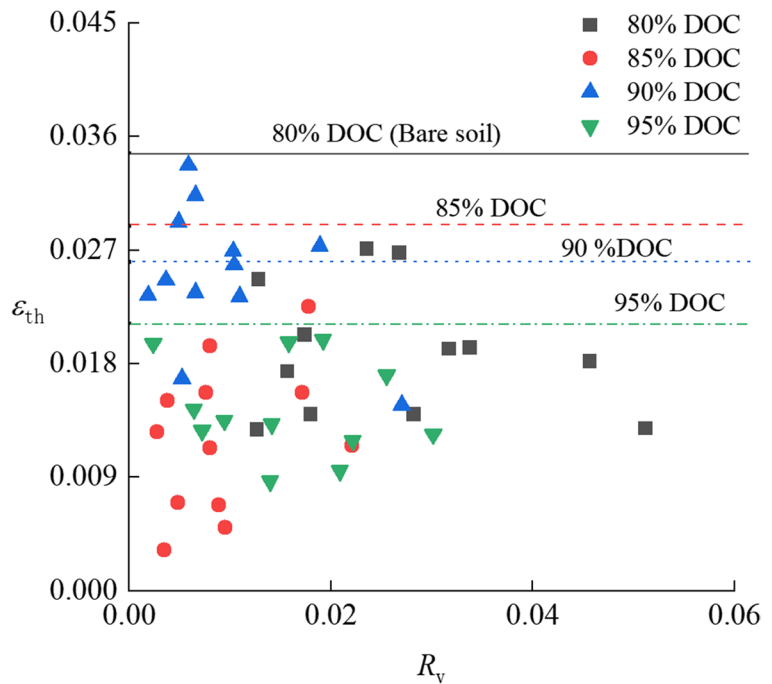


Fig. 7 Comparisons between measured and predicted gas permeability of rooted soil with different degrees of compaction: **a** 80% degree of compaction; **b** 85% degree of compaction; **c** 90% degree of compaction; **d** 95% degree of compaction

(RMSE represents root mean square error between the measured and predicted k_a . The subscript b and r denote bare and rooted soil, respectively. The predicted k_a becomes close to the measured value as RMSE decreases.)

Fig. 8 Effects of root volume ratio on the percolation threshold (ε_{th}) with different degrees of compaction (horizontal line and data point represent ε_{th} of bare and rooted soils, respectively)



increased (i.e., at a lower ε) under the same dry density, because more macropores were generated during soil compaction at a water content less than the optimal level, which increased the velocity of advective gas flow (Nguyen et al. 2021; Zhan et al. 2014). Thus, the macropores control the magnitude of k_a rather than that of ε . In contrast, D_p is mainly affected by ε and tortuosity, because the diffusive flow velocity of gas is independent of soil pore size for a given ambient condition, as described by the kinetic theory of molecules (Feng et al. 2022a).

Effects of roots on k_a and D_p decreased at a higher water content. It was because that the gas phase was discontinuous in soils with a high water content (i.e., $\varepsilon \leq 0.1$), which limited gas diffusion through the soil, as gas diffusion through a water phase is approximately four orders of magnitude lower than that through a gas phase (Feng et al. 2020). Moreover, at a higher degree of compaction, the bare soil exhibited a slight increase in D_p for a given ε (Fig. 6a–d), although a smaller pore size was observed in the soil with a higher degree of compaction (Fig. 4a and c), consistent with D_p measurement results for soils with different degrees of compaction obtained by Hamamoto et al. (2011). It was because the lower degree of saturation of soil with a higher degree of compaction (i.e., a lower porosity) under the same ε led to such

soil having a lower probability of pore blockage by water obstacles than soil with a lower degree of compaction. In contrast, the k_a of the bare soil decreased as the degree of compaction increased (Fig. 3a–d), owing to the reduction in soil pore size.

Different root characteristics, soil heterogeneity, root growth or decay, climate condition and plant species can increase the complexity of root–soil interactions. For an instance, the present study found that k_a of rooted soil was larger than that of bare soil under the degree of compaction of 95% (Fig. 3d). In contrast, Ni and Ng (2019) reported that roots of Bermuda grass reduced k_a of soil with a degree of compaction of 95%. This was likely because Ni and Ng (2019) planted Bermuda grass in a lab with good control of relative humidity, temperature and light, while the grass in the present study was left outdoors where the climate conditions were worse than that in the plant room. Thus, grass roots might decay herein, leading to an increase in k_a . It should be noted that effects of roots on k_a and D_p of soil would change with plant age. Further effort is needed to study long-term effects (e.g., grass grows for one or several years) of roots on k_a and D_p of soil. In addition, the tested soil exhibited no pronounced crack, which is attributable to its granular nature (it was a silty sand) and the relatively short test period. However, over the

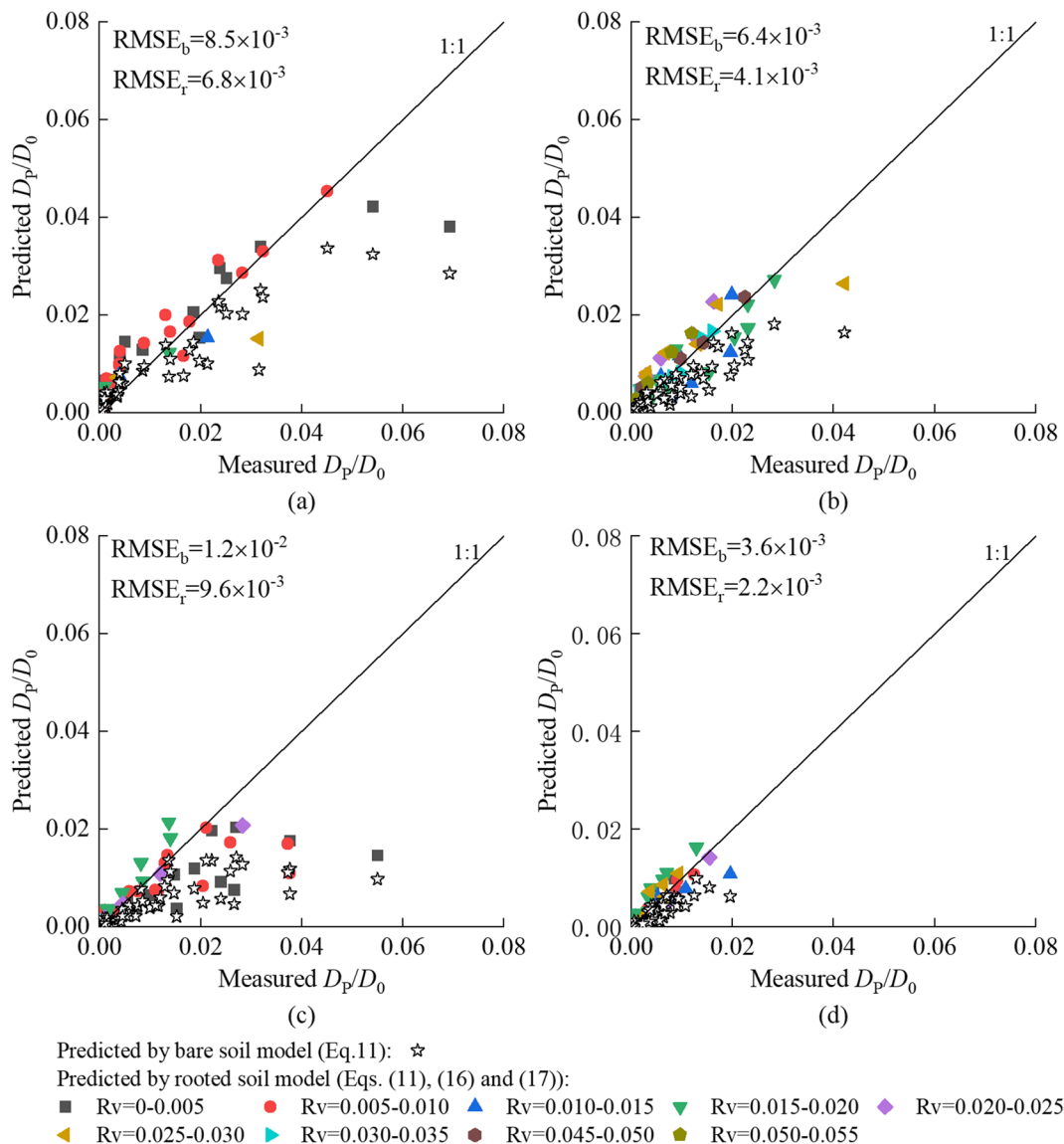


Fig. 9 Comparisons between measured and predicted relative gas diffusion coefficient of soil with different degrees of compaction: **a** 80% degree of compaction; **b** 85% degree of compaction; **c** 90% degree of compaction; **d** 95% degree of com-

paction (R^2 represents the fitting coefficients describing the goodness of best fitting through 1:1 line to the dataset of bare soil and rooted soil)

long term, roots may recolonise pores generated by decayed roots (Rasse and Smucker 1998). Moreover, cracks develop in fine-grained soil under long-term wetting–drying cycles (e.g., clay; Li et al. 2022). In addition, various root characteristics can induce soil microstructure heterogeneity, resulting in the anisotropy of k_a and D_p . This is evidenced by the noticeable scatter of k_a even under the same R_v (i.e., datasets

with ε of 0.10–0.12 and R_v of ~0.01 in Fig. S1c). All of the above factors attest to the complicated influence of roots on k_a and D_p . Further research is needed to investigate other factors influencing the k_a and D_p of rooted soil, such as plant species (shrubs, trees, or mixed plantings), plant age, long-term wetting–drying cycles, and soil anisotropy.

Conclusions

This study investigated the effects of the root characteristics of *C. dactylon* on the k_a and D_p of soil with various degrees of compaction. The correlations between root characteristic parameters and the k_a and D_p of rooted soil were evaluated via grey relational analysis, while soil microstructure was analysed via scanning electron microscopy and mercury intrusion porosimetry. Furthermore, empirical predictive models of k_a and D_p in unsaturated soil were established that consider the influence of root characteristics.

Grey relational analysis showed that compared with other frequently used root characteristic parameters, including root area index, root area ratio, R_b , and root length density, R_v had a greater effect on k_a and D_p . Overall, the rooted and bare soils with an 80% degree of compaction exhibited comparable k_a or D_p , regardless of R_v and ε . This was because the relatively large pore size of soil matrix with an 80% degree of compaction – as evidenced by the measured pore size distribution – limited the influence of roots on the soil microstructure. As degree of compaction increased, the k_a and D_p of roots generally increased. The differences in k_a or D_p between the rooted and bare soils became more significant as ε increased, owing to the presence of arc-shape macropores at the root–soil interface, as revealed by scanning electron microscopy. Thus, the roots also reduced the percolation threshold (ε_{th}) for gas diffusion. In highly compacted soil (an 85–95% degree of compaction), root occupancy increased k_a by more than one order of magnitude but increased D_p by a much lower degree. Moreover, roots increased k_a within the entire considered ε range but increased D_p only at an $\varepsilon \geq 0.10$, because k_a was controlled by macropores, while D_p was controlled by ε . In soil with a degree of compaction of 85–95%, the root-induced increase in k_a and D_p decreased with increasing R_v , because root occupancy reduced soil porosity.

Author contributions SF proposed the study conception and designed experiments; SFH and NKZ prepared the samples; SF, FQC and CWWN analyzed the data; SF, XQ, CWWN and SFH wrote and revised the manuscript. All authors read and approved the final manuscript.

Funding This project was supported by the National Natural Science Foundation of China (grant nos. 52178320 and 42177120), the National Key Research and Development

Program of China (2019YFC1806003), and the Chongqing Key Laboratory of Geomechanics & Geoenvironment Protection (LQ21KFJJ13).

Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

- Ali W, Yang MX, Long Q, Hussain S, Chen JZ, Clay D, He YB (2022) Different fall/winter cover crop root patterns induce contrasting red soil (Ultisols) mechanical resistance through aggregate properties. *Plant Soil* 477(1–2):461–474. <https://doi.org/10.1007/s11104-022-05430-4>
- Arsenault JL, Pouleur S, Messier C, Guay R (1995) WinRHIZOTM, a root-measuring system with a unique overlap correction method. *Hortscience* 30(4):906. <https://doi.org/10.21273/hortsci.30.4.906>
- ASTM (2007) D422: Standard test method for particle-size analysis of soils. ASTM International, West Conshohocken. <https://doi.org/10.1520/D0422-63R07>
- ASTM (2010) D698: Standard test method for laboratory compaction characteristics of soil using standard effort. ASTM International, West Conshohocken. <https://doi.org/10.1520/D0698-12R21>
- ASTM (2014) D854: Standard test method for specific gravity of soil solids by water pycnometer. ASTM International, West Conshohocken. <https://doi.org/10.1520/D0854-14>
- ASTM (2017a) D2487-17: Standard practice for classification of soils for engineering purposes (Unified Soil Classification System). ASTM International, West Conshohocken. <https://doi.org/10.1520/D2487-17>
- ASTM (2017b) D4318: Standard test methods for liquid limit, plastic limit, and plasticity index of soils. ASTM International, West Conshohocken. <https://doi.org/10.1520/D4318-17>
- ASTM (2018) D4404-18. Determination of Pore Volume and Pore Volume Distribution of Soil and Rock by Mercury Intrusion Porosimetry. ASTM standard. American for Testing and Materials. ASTM International, West Conshohocken. <https://doi.org/10.1520/D4404-18>
- Ball BC, O'Sullivan MF, Hunter R (1988) Gas diffusion, fluid flow and derived pore continuity indices in relation to vehicle traffic and tillage. *J Soil Sci* 39(3):327–339. <https://doi.org/10.1111/j.1365-2389.1988.tb01219.x>
- Berisso FE, Schjønning P, Keller T, Lamandé M, Simojoki A, Iversen BV, Alakukku L, Forkman J (2013) Gas transport and subsoil pore characteristics: Anisotropy and long-term effects of compaction. *Geoderma* 195:184–191. <https://doi.org/10.1016/j.geoderma.2012.12.002>
- Bian RX, Shi W, Chai XL, Sun YJ (2020) Effects of plant radial oxygen loss on methane oxidation in landfill cover soil: A simulative study. *Waste Manage* 102:56–64. <https://doi.org/10.1016/j.wasman.2019.10.033>

- Boldrin D, Leung AK, Bengough AG (2017) Correlating hydrologic reinforcement of vegetated soil with plant traits during establishment of woody perennials. *Plant Soil* 416:437–451. <https://doi.org/10.1007/s11104-017-3211-3>
- Bouazza A, Vangpaisal T (2003) An apparatus to measure gas permeability of geosynthetic clay liners. *Geotext Geomembr* 21(2):85–101. <https://doi.org/10.1016/j.geotexmem.2003.10.001>
- Carminati A, Vetterlein D, Koebnick N, Blaser S, Weller U, Vogel HJ (2013) Do roots mind the gap? *Plant Soil* 367(1–2):651–661. <https://doi.org/10.1007/s11104-012-1496-9>
- Chen G, Weil RR, Hill RL (2014) Effects of compaction and cover crops on soil least limiting water range and air permeability. *Soil Tillage Res* 136:61–69. <https://doi.org/10.1016/j.still.2013.09.004>
- Chen C, Zhang D, Zhang J (2017) Influence of stress and water content on air permeability of intact loess. *Can Geotech J* 54(9):1221–1230. <https://doi.org/10.1139/cgj-2016-0186>
- Chen FQ, Zhao NK, Feng S, Liu HW, Liu YC (2020) Effects of biochar content on gas diffusion coefficient of soil with different compactness and air contents. *Environ Sci Pollut Res* 27(17):21497–21505. <https://doi.org/10.1007/s11356-020-08594-7>
- Chen R, Huang JW, Zhou C, Ping Y, Chen ZK (2021) A new simple and low-cost air permeameter for unsaturated soils. *Soil Tillage Res* 213:105083. <https://doi.org/10.1016/j.still.2021.105083>
- Feng S, Leung AK, Liu HW, Ng CWW (2020) Modelling microbial growth and biomass accumulation during methane oxidation in unsaturated soil. *Can Geotech J* 57(2):189–204. <https://doi.org/10.1139/cgj-2018-0147>
- Feng S, Sun JX, Liu HW, Zhan LT (2022b) A new method and instrument of measuring in-situ gas diffusion coefficient and gas coefficient of permeability of unsaturated soil. *J Geotech Geoenviron* 149(7):04023041
- Feng S, Huang SF, Jiang JL, Zhan LT, Li GY, Guan RQ, Guo HW, Liu HW (2022a) Effects of pore-size distribution on gas diffusion coefficient and gas permeability of compacted manufactured sand tailings-bentonite mixtures. *J Geotech Geoenviron* (in press)
- Gilman EF, Leone IA, Flower FB (1982) Influence of soil gas contamination on tree root growth. *Plant Soil* 65:3–10. <https://doi.org/10.1007/BF02376797>
- Glauz RD, Rolston DE (1989) Optimal design of two-chamber, gas diffusion cells. *Soil Sci Soc Am J* 53(6):1619–1624. <https://doi.org/10.2136/sssaj1989.03615995005300060001x>
- Hamamoto S, Moldrup P, Kawamoto K, Komatsu T (2009) Effect of particle size and soil compaction on gas transport parameters in variably saturated, sandy soils. *Vadose Zone J* 8(4):986–995. <https://doi.org/10.1016/j.wasman.2011.07.008>
- Hamamoto S, Moldrup P, Kawamoto K, Wickramarachchi PN, Nagamori M, Komatsu T (2011) Extreme compaction effects on gas transport parameters and estimated climate gas exchange for a landfill final cover soil. *J Geotech Geoenviron* 137(7):653–662. [https://doi.org/10.1061/\(ASCE\)GT.1943-5606.0000459](https://doi.org/10.1061/(ASCE)GT.1943-5606.0000459)
- Hamamoto S, Moldrup P, Kawamoto K, Komatsu T (2012) Organic matter fraction dependent model for predicting the gas diffusion coefficient in variably saturated soils. *Vadose Zone J* 11(1):1–9. <https://doi.org/10.2136/vzj2011.0065>
- Hu X, Li XY, Wang P, Liu Y, Wu XC, Li ZC, Zhao YD, Cheng YQ, Guo LL, Lyu YL, Liu LY (2019) Influence of enclosure on CT-measured soil macropores and root architecture in a shrub-encroached grassland in northern China. *Soil Tillage Res* 187:21–30. <https://doi.org/10.1016/j.still.2018.10.020>
- Ikedo J, Nakamura T (1996) Mathematical model of lateral oxygen diffusion and respiration of roots for estimating the diffusion coefficient and affinity to oxygen. *Soil Sci Plant Nutr* 42(4):691–699. <https://doi.org/10.1080/00380768.1996.10416616>
- Jotisankasa A, Sirirattanachai T (2017) Effects of grass roots on soil-water retention curve and permeability function. *Can Geotech J* 54(11):1612–1622. <https://doi.org/10.1139/cgj-2016-0281>
- Kamchoom V, Leung AK, Boldrin D, Sakolpanya T, Wu Z, Likitlersuang S (2022a) Shearing behaviour of vegetated soils with growing and decaying roots. *Can Geotech J* 59(12):2067–2084. <https://doi.org/10.1139/cgj-2021-0695>
- Kamchoom V, Boldrin D, Leung AK, Sookkrajang C, Likitlersuang S (2022b) Biomechanical properties of the growing and decaying roots of *Cynodon dactylon*. *Plant Soil* 471:193–210. <https://doi.org/10.1007/s11104-021-05207-1>
- Kamchoom V, Chen XW, Leung AK, Sakolpanya T, Srinil C (2023) Dynamic changes in cellulose content and biomechanical properties of mycorrhizal roots during growth and decay. *Plant Soil* 1–17. <https://doi.org/10.1007/s11104-023-06103-6>
- Kautz T, Stumm C, Kösters R, Köpke U (2010) Effects of perennial fodder crops on soil structure in agricultural headlands. *J Plant Nutr Soil Sci* 173(4):490–501. <https://doi.org/10.1002/jpln.200900216>
- Kiuru P, Palviainen M, Grönholm T, Raivonen M, Kohl L, Gauci V, Urzainki I, Laurén A (2022) Peat macropore networks—new insights into episodic and hotspot methane emission. *Biogeosciences* 19(7):1959–1977. <https://doi.org/10.5194/bg-19-1959-2022>
- Leung AK, Garg A, Ng CWW (2015) Effects of plant roots on soil-water retention and induced suction in vegetated soil. *Eng Geol* 193:183–197. <https://doi.org/10.1016/j.enggeo.2015.04.017>
- Li GY, Zhan LT, Zhang S, Feng S, Zhang ZH (2022) A dual-porosity model for coupled rainwater and landfill gas transport through capillary barrier covers. *Comput Geotech* 151:104966. <https://doi.org/10.1016/j.compgeo.2022.104966>
- Lipiec J, Horn R, Pietrusiewicz J, Siczek A (2012) Effects of soil compaction on root elongation and anatomy of different cereal plant species. *Soil Tillage Res* 121:74–81. <https://doi.org/10.1016/j.still.2012.01.013>
- Liu HW, Feng S (2022) Effects of nano-biochar of different particle sizes on the shrinkage properties of kaolin. *J Soils Sediments* 22:1511–1520. <https://doi.org/10.1007/s11368-022-03161-8>
- Liu XP, Zhang WJ, Wang XY, Cai YJ, Chang JG (2015) Root-soil air gap and resistance to water flow at the soil-root interface of *Robinia pseudoacacia*. *Tree Physiol*

- 35(12):1343–1355. <https://doi.org/10.1093/treephys/tpv075>
- Liu HW, Feng S, Ng CWW (2018) Analytical solutions of pore-water pressure distributions in a vegetated multi-layered slope considering the effects of roots on water permeability. *Comput Geotech* 102:252–261. <https://doi.org/10.1016/j.compgeo.2018.06.003>
- Luo ZT, Niu JZ, Zhang L, Chen XW, Zhang W, Xie BY, Du J, Zhu ZJ, Wu SS, Li X (2019) Roots-enhanced preferential flows in deciduous and coniferous forest soils revealed by dual-tracer experiments. *J Environ Qual* 48(1):136–146. <https://doi.org/10.2134/jeq2018.03.0091>
- Mäki M, Aaltonen H, Heinonsalo J, Hellén H, Pumpanen J, Bäck J (2019) Boreal forest soil is a significant and diverse source of volatile organic compounds. *Plant Soil* 441:89–110. <https://doi.org/10.1007/s11104-019-04092-z>
- Moldrup P, Olesen T, Gamst J, Schjønning P, Yamaguchi T, Rolston DE (2000) Predicting the gas diffusion coefficient in repacked soil: Water-induced linear reduction model. *Soil Sci Soc Am J* 64(5):1588–1594. <https://doi.org/10.2136/sssaj2000.6451588x>
- Ng CWW (2017) Atmosphere-plant-soil interactions: Theories and mechanisms. *Chin J Geotech Eng* 39(1):1–47 (In Chinese)
- Ng CWW, Liu HW, Feng S (2015) Analytical solutions for calculating pore-water pressure in an infinite unsaturated slope with different root architectures. *Can Geotech J* 52(12):1981–1992. <https://doi.org/10.1139/cgj-2015-0001>
- Ng CWW, Garg A, Leung AK, Hau BCH (2016) Relationships between leaf and root area indices and soil suction induced during drying-wetting cycles. *Ecol Eng* 91:113–118. <https://doi.org/10.1016/j.ecoleng.2016.02.005>
- Ng CWW, Guo HW, Ni JJ, Chen R, Xue Q, Feng Y, Chen ZK, Feng S, Zhang Q (2022a) Long-term field performance of non-vegetated and vegetated three-layer landfill cover systems using construction waste without geomembrane. *Géotechnique*, in press. <https://doi.org/10.1680/jgeot.21.00238>
- Ng CWW, Guo HW, Ni JJ, Zhang Q, Chen ZK (2022b) Effects of soil-plant-biochar interactions on water retention and slope stability under various rainfall patterns. *Landslides* 19:1379–1390. <https://doi.org/10.1007/s10346-022-01874-y>
- Nguyen V, Pineda JA, Romero E, Sheng D (2021) Influence of soil microstructure on air permeability in compacted clay. *Géotechnique* 71(5):373–391. <https://doi.org/10.1680/jgeot.18.P.186>
- Ni JJ, Ng CWW (2019) Long-term effects of grass roots on gas permeability in unsaturated simulated landfill covers. *Sci Total Environ* 666:680–684. <https://doi.org/10.1016/j.scitotenv.2019.02.248>
- Nobel PS, Cui M (1992) Hydraulic conductances of the soil, the root-soil air gap, and the root: changes for desert succulents in drying soil. *J Exp Bot* 43(248):319–326. <https://doi.org/10.1093/jxb/43.3.319>
- Pang W, Crow WT, Luc JE, Mcorley R, Kruse JK (2011) Comparison of water displacement and WinRHIZO software for plant root parameter assessment. *Plant Dis* 95(10):1308–1310. <https://doi.org/10.1094/PDIS-01-11-0026>
- Rasse DP, Smucker AJM (1998) Root recolonization of previous root channels in corn and alfalfa rotations. *Plant Soil* 204:203–212. <https://doi.org/10.1023/A:1004343122448>
- Reible DD, Shair FH (1982) A technique for the measurement of gaseous diffusion in porous media. *J Soil Sci* 33(2):165–174. <https://doi.org/10.1111/j.1365-2389.1982.tb01756.x>
- Ren LD, Nest TV, Ruyschaert G, D'Hose T, Cornelis WM (2019) Short-term effects of cover crops and tillage methods on soil physical properties and maize growth in a sandy loam soil. *Soil Tillage Res* 192:76–86. <https://doi.org/10.1016/j.still.2019.04.026>
- Resurreccion AC, Moldrup P, Kawamoto K, Yoshikawa S, Rolston DE, Komatsu T (2008) Variable pore connectivity factor model for gas diffusivity in unsaturated, aggregated soil. *Vadose Zone J* 7(2):397–405. <https://doi.org/10.2136/vzj2007.0058>
- Rouf MA, Singh RM, Bouazza A, Rowe RK, Gates WP (2016) Gas permeability of partially hydrated geosynthetic clay liner under two stress conditions. *Environ Geotech* 3(5):325–333. <https://doi.org/10.1680/envgeo.14.00009>
- Schaffer B, Stauber M, Muller R, Schulin R (2007) Changes in the macro-pore structure of restored soil caused by compaction beneath heavy agricultural machinery: a morphometric study. *Eur J Soil Sci* 58:1062–1073. <https://doi.org/10.1111/j.1365-2389.2007.00886.x>
- Schjønning P, Thomsen IK, Moldrup P, Christensen BT (2003) Linking soil microbial activity to water and air-phase contents and diffusivities. *Soil Sci Soc Am J* 67(1):156–165. <https://doi.org/10.2136/sssaj2003.1560>
- Song L, Li JH, Zhou T, Fredlund DG (2017) Experimental study on unsaturated hydraulic properties of vegetated soil. *Ecol Eng* 103:207–216. <https://doi.org/10.1016/j.ecoleng.2017.04.013>
- Tang AM, Cui YJ, Richard G, Défossez P (2011) A study on the air permeability as affected by compression of three French soils. *Geoderma* 162(1–2):171–181. <https://doi.org/10.1016/j.geoderma.2011.01.019>
- Townsend-Small A, Czimczik CI (2010) Carbon sequestration and greenhouse gas emissions in urban turf. *Geophys Res Lett* 37(2):195–205. <https://doi.org/10.1029/2009GL041675>
- Unger PW, Kaspar TC (1994) Soil compaction and root-growth – a review. *Agron J* 86(5):759–766. <https://doi.org/10.2134/agronj1994.00021962008600050004x>
- Uteau D, Pagenkemper SK, Peth S, Horn R (2013) Root and time dependent soil structure formation and its influence on gas transport in the subsoil. *Soil Tillage Res* 132:69–76. <https://doi.org/10.1016/j.still.2013.05.001>
- Xu M, Zhu Q, Wu J, He Y, Yang G, Zhang X, Li L, Yu X, Peng H, Wang L (2018) Grey relational analysis for evaluating the effects of different rates of wine lees-derived biochar application on a plant-soil system with multi-metal contamination. *Environ Sci Pollut Res* 25(7):6990–7001. <https://doi.org/10.1007/s11356-017-1048-1>
- Yue L, Wang Y, Wang L, Yao S, Cong C, Ren L, Zhang B (2021) Impacts of soil compaction and historical soybean variety growth on soil macropore structure. *Soil Tillage Res* 214:105166. <https://doi.org/10.1016/j.still.2021.105166>
- Zhan TLT, Yang YB, Chen R, Ng CWW, Chen YM (2014) Influence of clod size and water content on gas permeability of a compacted loess. *Can Geotech J* 51(12):1468–1474. <https://doi.org/10.1139/cgj-2014-0126>
- Zhan LT, Qiu QW, Xu WJ, Chen YM (2016) Field measurement of gas permeability of compacted loess used as an

- earthen final cover for a municipal solid waste landfill. *J Zhejiang Univ-Sci A* 17(7):541–552. <https://doi.org/10.1631/jzus.A1600245>
- Zhan LT, Wu T, Feng S, Li GY, He HJ, Lan JW, Chen YM (2020) Full-scale experimental study of methane emission in a loess-gravel capillary barrier cover under different seasons. *Waste Manage* 107:54–65. <https://doi.org/10.1016/j.wasman.2020.03.026>
- Zhan LT, Ni JQ, Feng S, Kong LG, Feng T (2022) Saturated hydraulic conductivity of compacted steel slag-bentonite mixtures—a potential hydraulic barrier material. *Waste Manage* 144:349–356. <https://doi.org/10.1016/j.wasman.2022.04.004>
- Zhang ZB, Peng XH (2021) Bio-tillage: A new perspective for sustainable agriculture. *Soil Tillage Res* 206:104844. <https://doi.org/10.1016/j.still.2020.104844>
- Zhang ZB, Yan L, Wang YK, Ruan RJ, Xiong P, Peng XH (2022) Bio-tillage improves soil physical properties and maize growth in a compacted Vertisol by cover crops. *Soil Sci Soc Am J* 86(2):324–337. <https://doi.org/10.1002/saj2.20368>
- Zhu J, Leung AK, Wang Y (2022a) Modelling root–soil mechanical interaction considering root pull-out and breakage failure modes. *Plant Soil* 480:675–701. <https://doi.org/10.1007/s11104-022-05613-z>
- Zhu ZW, Feng SJ, Chen HX, Chen ZL, Ding XH, Peng CH (2022b) Approximate analytical model for transient transport and oxygen-limited biodegradation of vapor-phase petroleum hydrocarbon compound in soil. *Chemosphere* 300:134522. <https://doi.org/10.1016/j.chemosphere.2022.134522>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.